SCTR's PUNE INSTITUTE OF COMPUTER TECHNOLOGY, PUNE

ELECTRONICS AND TELECOMMUNICATION ENGINEERING

"Dexterous"

Guide: Prof . Lalit Patil

By:

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Introduction

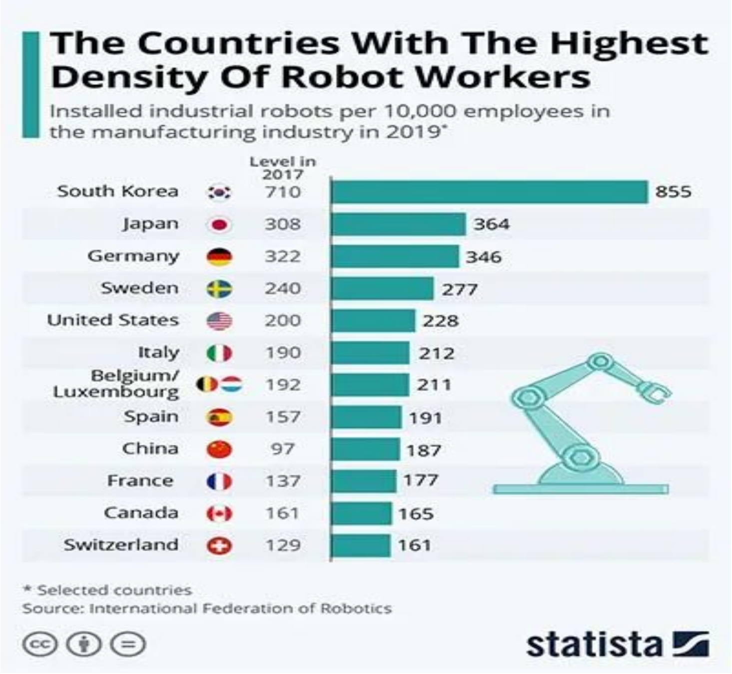
1. Challenge and Innovation :
   a. Crafting a 6-DOF robotic arm with ESP32 and Raspberry Pi 3 is a thrilling challenge.
   b. Integrating these technologies adds complexity and sparks innovation.
2. Skill Development :
   a. Involves honing mechanical, electronic, and coding skills.
   b. Opportunities to master ESP32 and Raspberry Pi 3 intricacies.
3. Dynamic Control :
   a. Achieving a 6-DOF setup allows for intricate control and dynamic response.
   b. Witnessing precise arm responses becomes a tangible achievement.
4. Future Automation Insight :
   a. Offers deep insight into future automation with ESP32 and Raspberry Pi 3.
   b. Gaining practical knowledge positions you as a contributor to future advancements.
5. Creating a 6-DOF robotic arm with locomotion using ESP32 and Raspberry Pi 3 isn't just a project; it's a transformative experience propelling you into cutting-edge technology.

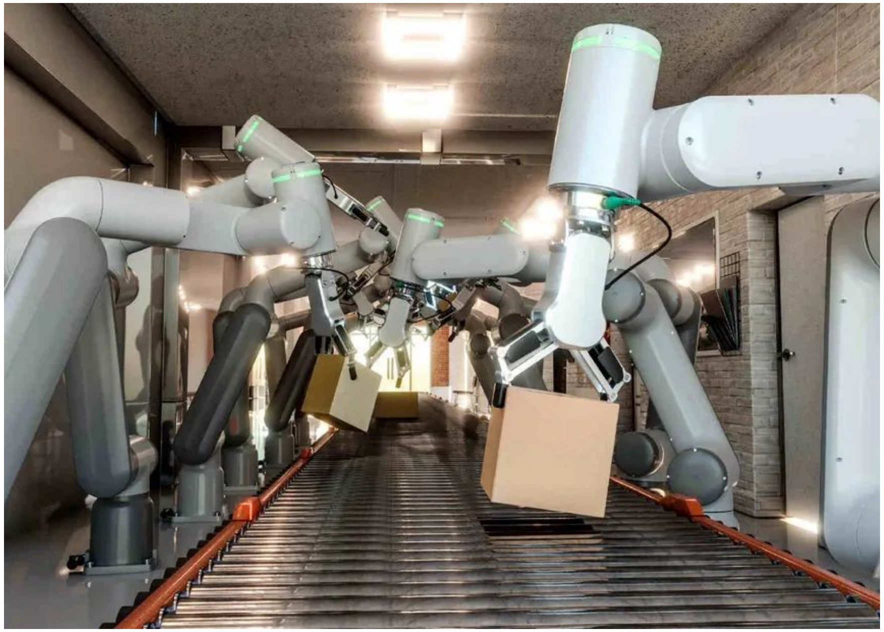
Literature survey

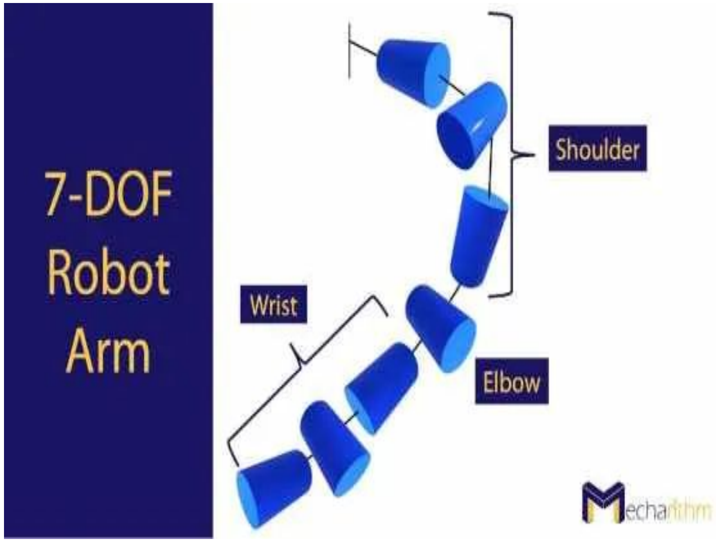
1. Introduction :
   a. Interest in versatile robotic arms with locomotion, merging agility and mobility.
   b. Focus on 6-DOF arms using ESP32 and Raspberry Pi 3.
2. Study 1 : "Advanced Robotic Arm Control via IoT with ESP32 and Raspberry Pi 3" (2020)
   a. Features: 6 DOFs, ESP32, Raspberry Pi 3, IoT connectivity.
   b. Insight: Demonstrates advanced control capabilities through IoT.
   c. Source: Academia.edu
3. Study 2 : "Affordable 3D-Printed 6-DOF Robotic Arm with ESP32 and Raspberry Pi 3" (2021)
   a. Features: 6 DOFs, 3D-printed, Bluetooth control.
   b. Insight: Highlights affordability and flexibility in a 6-DOF context.
   c. Source: Hiwonder

Literature survey

1. Study 3 : "Integrating Locomotion with a 6-DOF Robotic Arm using ESP32 and Raspberry Pi 3" (2019)
   a. Features: 6 DOFs, mounted on mobile platform, synchronized control with Raspberry Pi 3.
   b. Insight: Successful integration of arm manipulation with mobile locomotion.
   c. Source: Research Hub
2. Key Trends and Findings : ESP32, Raspberry Pi 3 common for control, showcasing synergy. 3D printing vital for cost-effective 6-DOF fabrication. Trend towards integrating 6-DOF arms with locomotion.
3. Conclusion : Advances in 6-DOF arms with locomotion, using ESP32 and Raspberry Pi 3. Trends include advanced control, affordability, and arm-locomotion integration. Future research opportunities in refining controls and expanding applications.


The Countries With The Highest Density Of Robot Workers
Installed industrial robots per 10,000 employees in the manufacturing industry in 2019*
Level in 2017
South Korea 710 855
Japan 308 364
Germany 322 346
Sweden 240 277
United States 200 228
Italy 190 212
Belgium/ Luxembourg 192 211
Spain 157 191
China 97 187
France 137 177
Canada 161 165
Switzerland 129 161
* Selected countries
Source: International Federation of Robotics
statista

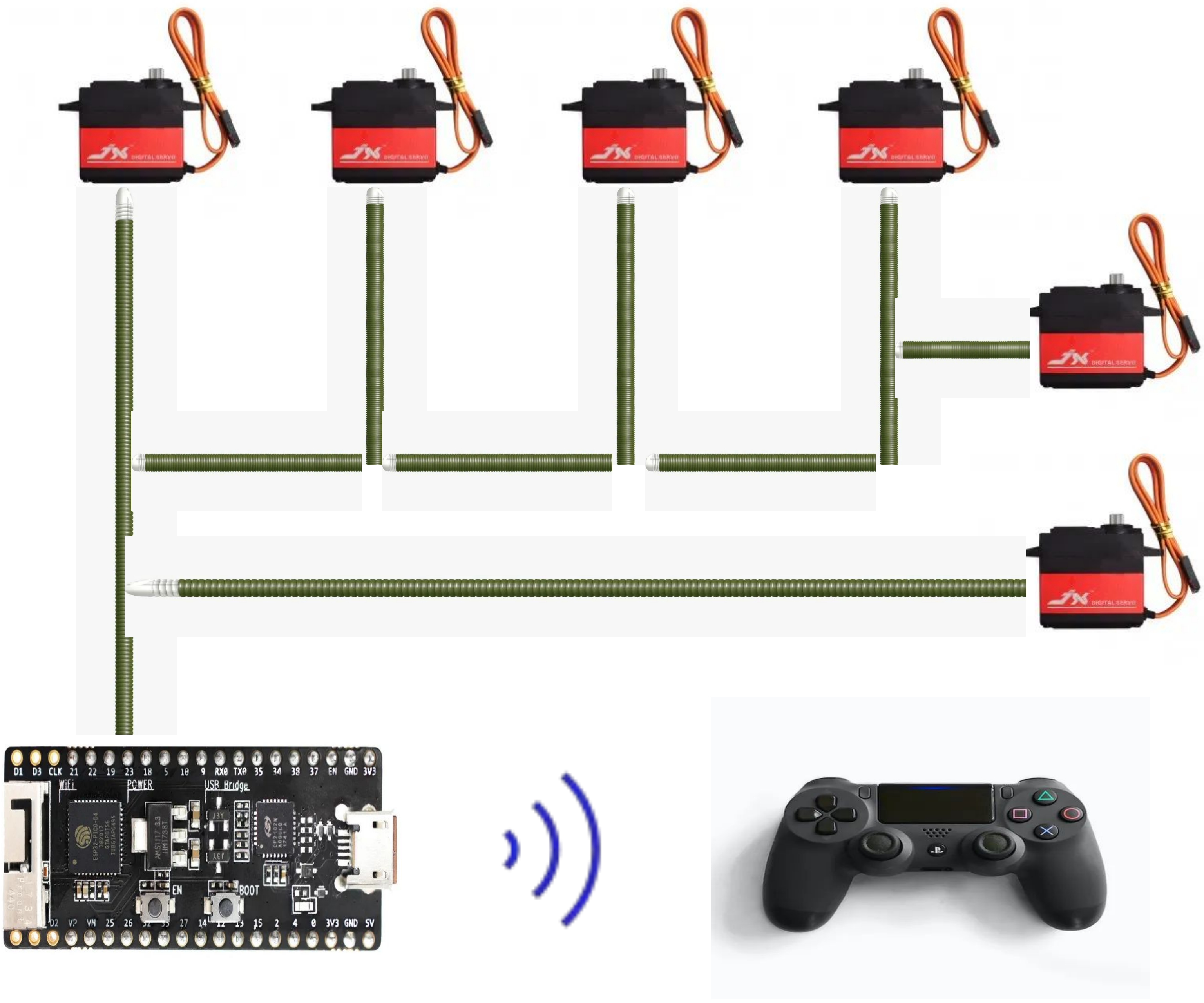



7-DOF Robot Arm
Shoulder
Wrist
Elbow

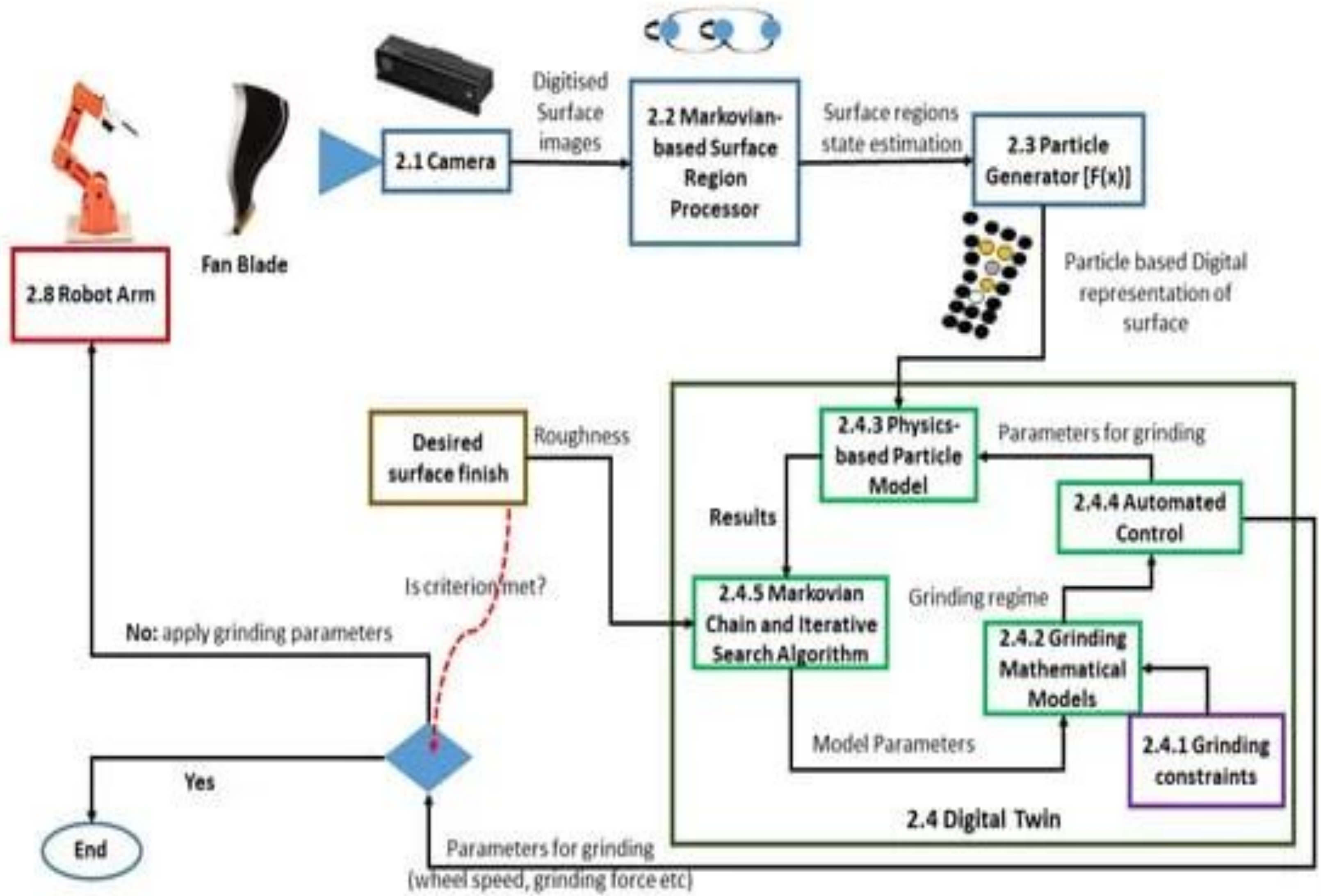
Objectives

1) Precise Control : Ensure smooth and accurate movement using ESP32 and servo motors.
2) Object Manipulation : Design the arm to grasp and manipulate objects of different shapes and sizes.
3) Wireless Communication : Utilize ESP32 for wireless control via Wi-Fi or Bluetooth, supporting joystick, web app, or smartphone interfaces.
4) User-Friendly Interface : Create an intuitive control interface for users of all skill levels.
5) Power Efficiency : Optimize battery life or power supply for extended operation.
6) Modular Design : Enable future upgrades with a modular platform for additional sensors, tools, or attachments.
7) Locomotion Integration : Implement effective movement capabilities for the robotic arm.
8) End-Effector Customization : Allow customization of end-effectors for various tasks.
9) Safety Measures : Integrate safety features for obstacle detection and secure operation.

Block Diagram /working model with animation

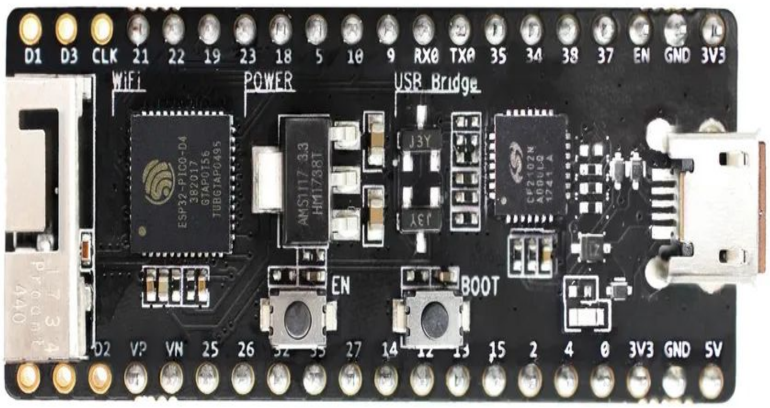

WiFi
POWER
USB Bridge
EN
BOOT

Industrial Application Diagram :



RESOURCES REQUIRED

- Hardware with specification



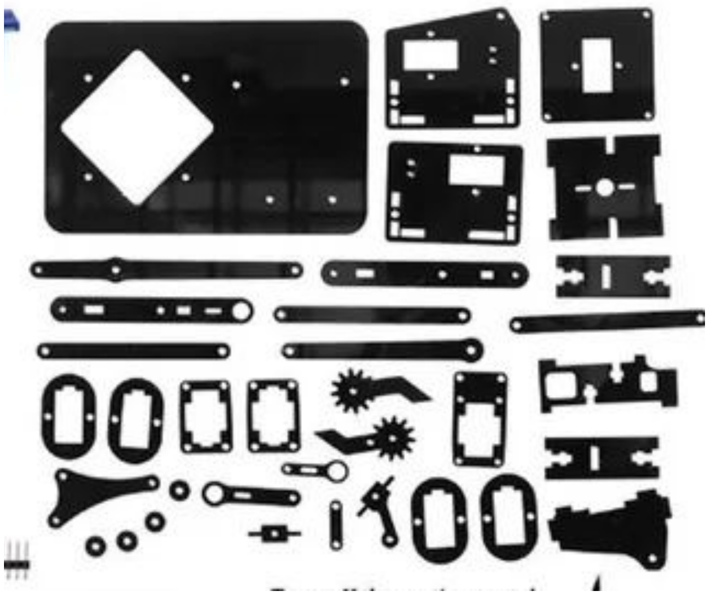
1) ESP-32 Module

2) Servo Motors - MG995

3) Toy Motors and Wheels

4) Jumper Cables

5) PS-4 Controller



6) 3D Prints or Laser Cut Material

7) PCB Breakout Board

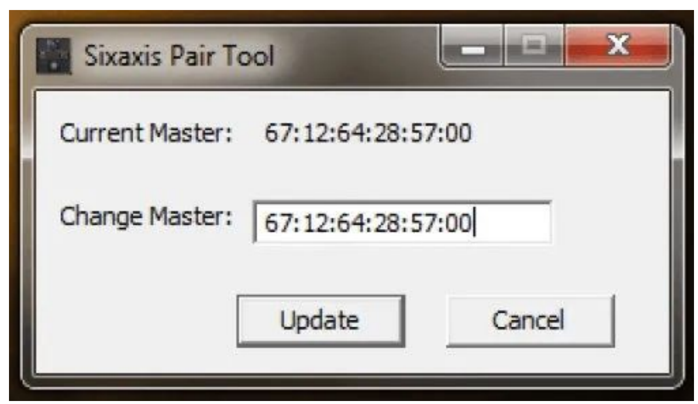
RESOURCES REQUIRED

- Software with version

1) Fusion 360

2) KiCAD / Altium Designer

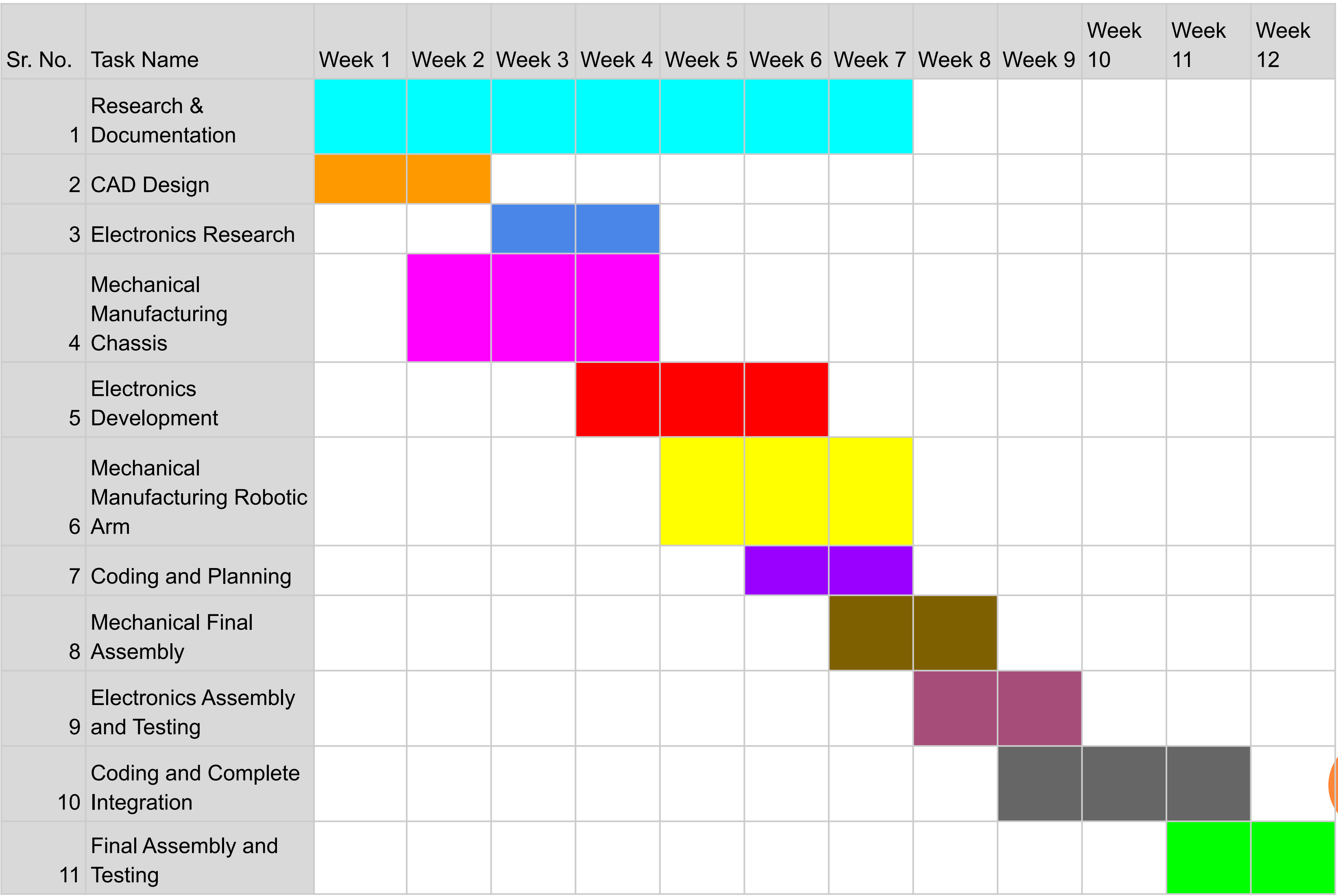
3) Arduino IDE 2.2.1



4) SixAxisPairTool

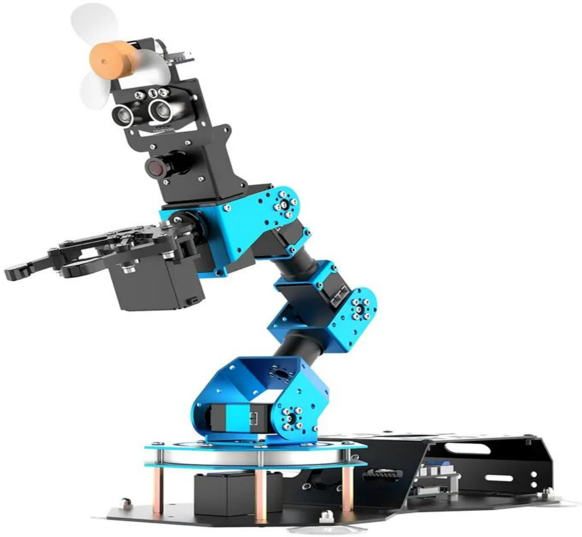
Gantt Chart

PROJECT SCHEDULE



References

1. Mohd Ashiq Kamaril Yusoff's, Reza Ezuan Samin, Babul Salam Ksm Kader Ibrahim, "Wireless Mobile Robotic Arm", ELSEVIER, ResearchGate, December 2012, 1072-1078.
2. https://iotdesignpro.com/articles/iot-based-robotic-arm-controlling-using-esp32
3. https://www.academia.edu/50180450/Design_and_implement_of_robotic_arm_and_control_of_moving_via_IoT_with_Arduino_ESP32
4. https://ozrobotics.com/shop/hiwonder-maxarm-robot-arm-powered-by-esp32-support-python-and-arduino-programming-standard-kit/
5. https://www.researchgate.net/publication/333886109_A_Cost-Efficient_Multipurpose_Service_Robot_using_Raspberry_Pi_and_6_DOF_Robotic_Arm
6. https://grabcad.com/library/robot-arm-83
7. https://docs.google.com/spreadsheets/d/1wXcu2AezT9oymwPx0rUag4GGjRIw_IFt9GSfBZX83_Y/edit#gid=0



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